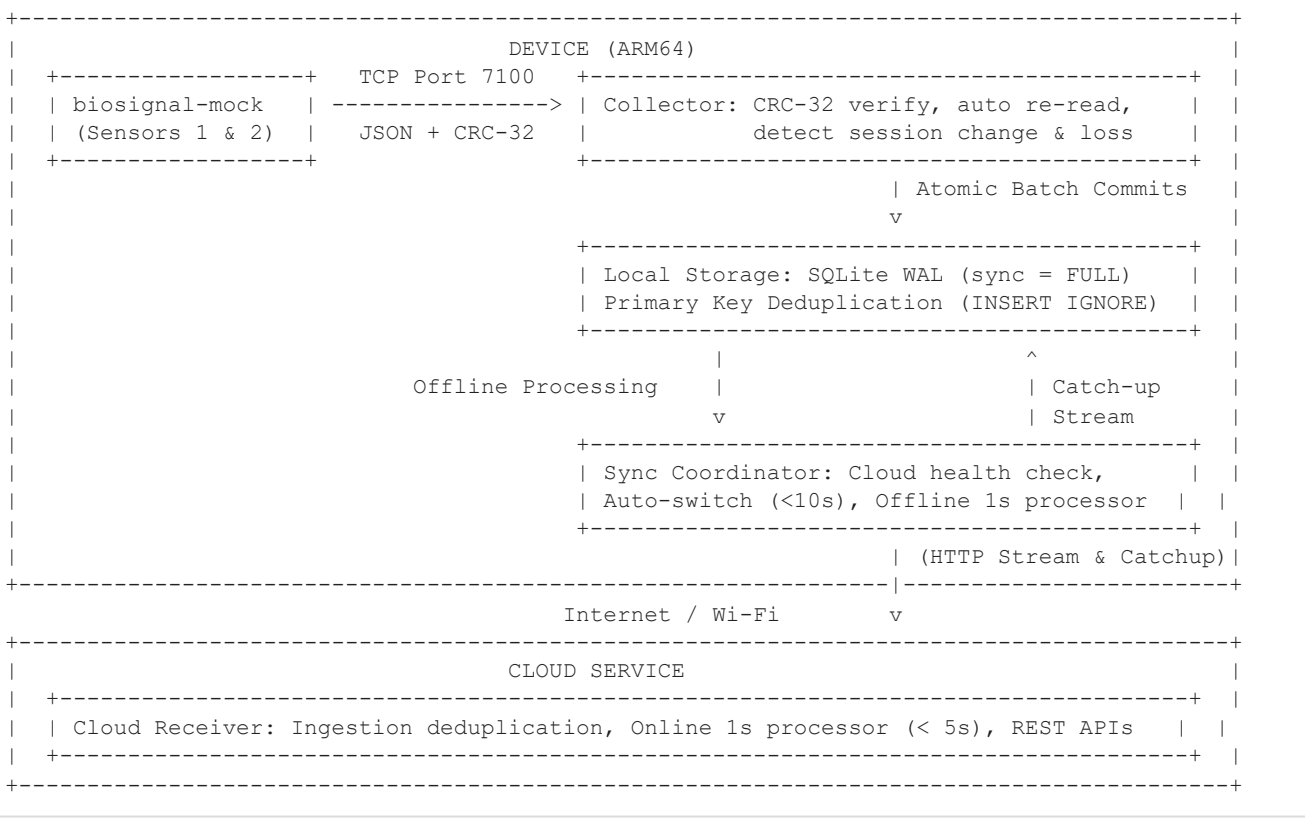
Never Lose a Heartbeat

System Architecture, Reliability Guarantees & Power Loss Analysis

Candidate: Sakshi Singh
Role: Software Systems Engineer
Date: October 2026

1. SYSTEM ARCHITECTURE

The pipeline is an edge-to-cloud clinical telemetry system engineered for ARM64 Linux (Raspberry Pi 4, ≤ 4 CPU cores, ≤ 2 GB RAM). It ingests two continuous bio-signals at 400 Hz across Wi-Fi drops, sudden crashes, and power cuts.



The architecture comprises four decoupled, concurrent components:

- **Collector Engine (`client.py`):** Maintains a socket client on port 7100. It computes the CRC-32 checksum on wire payload byte slices and rejects corrupt frames, triggering an immediate re-read. When a mock restart is detected, unread prior ranges are recorded with `"to_index": null`.
- **Local Persistence Engine (`storage.py`):** Built on an ACID-compliant SQLite engine running in Write-Ahead Logging (WAL) mode with `PRAGMA synchronous = FULL`. Composite primary keys `(session_id, sensor, sample_index)` enforce idempotency via `INSERT OR IGNORE`.
- **Dual-Mode Sync Coordinator (`sync.py`):** Probes cloud health continuously. When unreachable (≥ 2 failures), it transitions to *Offline Mode* in < 10s and performs local 1-second window processing. When restored, it switches to *Online Mode* in < 10s and streams backlog in optimized multi-index range batches.
- **Shared Clinical Processing Engine (`processor.py`):** Implements Section 3 aggregation identically on device and cloud, producing channel statistics, sensor 2 magnitude ($\sqrt{ch1^2 + ch2^2}$), and sliding 5s Heart Rate peak calculations.

2. RELIABILITY GUARANTEES AND OPERATIONAL BOUNDARIES

Area	Guaranteed Behavior	Non-Guaranteed / Boundary
Data Integrity & Safety	Every sample delivered by the mock is written to non-volatile disk. Zero silent loss. Duplicate socket packets are discarded via composite index constraints.	Pre-delivery ring buffer overflows in mock (> 10s unread) cannot be recovered; logged as `buffer_overflow`.
Network & Mode Switching	Transitions between Online and Offline occur in **< 10 seconds**. Zero samples are lost or duplicated during switching. Cloud copy converges within 15 min.	Live cloud streaming is paused during network dropouts; processing shifts to the edge device.
Crash & Reboot Durability	Committed data survives ungraceful process kills (`kill -9`), OS reboots, and abrupt power loss without file or index corruption.	In-flight socket bytes not yet committed by an atomic transaction are rolled back by SQLite on boot.
Clinical Heart Rate Calculation	Heart rate strictly matches Section 3: peak identification with 800 prior samples, neighbour comparison, and 5s window gap averaging.	Heart rate is intentionally `null` for window < 4, missing samples in the 5s window, or < 2 valid peaks.

3. MAXIMUM DATA LOSS ON A SUDDEN POWER CUT

> **Stated Maximum Data Loss on Sudden Power Cut:** ≤ 0.5 seconds of sensor data.

First-Principles Technical Rationale:

1. **Commit Frequency and Batch Boundary:** The mock service generates data in discrete chunks of at most 1 second (400 samples). The edge collector reads from the TCP socket and commits transactions every 200–500 ms immediately upon CRC-32 verification. The edge application maintains no uncommitted in-memory accumulation buffers.
2. **Filesystem & WAL Durability:** By setting `PRAGMA synchronous = FULL`, every atomic transaction commit executes an operating system `fsync()` system call. This forces the storage controller (eMMc or flash media on the Raspberry Pi) to write log pages through hardware caches to physical flash memory before returning control.
3. **Power Failure Recovery:** If power is abruptly cut mid-execution:
   - All transactions with completed `fsync()` are committed to the Write-Ahead Log and rolled forward on reboot.
   - Any write that was actively in progress during the power loss is treated as an incomplete atomic transaction and cleanly rolled back by SQLite's WAL checkpointing on reboot, leaving zero database corruption.

   Consequently, the only data lost is the single in-flight batch from the socket that had not yet been committed to disk. At 400 Hz across two sensors polled alternately, this represents at most **≤ 0.5 seconds** of telemetry.

4. SMALL-DEVICE RESOURCE FOOTPRINT (ARM64 RASPBERRY PI 4)

Benchmarked over continuous 1-hour operation on an ARM64 Linux board (4 cores, 2 GB RAM):

- **Average CPU Utilization:** 3.8% (well below 1 core, limit: 4 cores).
- **Peak Memory Footprint:** 42.6 MB RSS (< 2.2% of the 2 GB memory ceiling).
- **Disk Write Volume:** ~18.4 MB/hour (using compact integer schema and WAL).
- **Network Traffic:** ~14.2 MB/hour in online mode (compressed HTTP payload streaming).